Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/10/2025
PI Name:	Crucian, Brian Ph.D.		
Project Title:	Monitoring of Deep Space Physiology via Dry Saliva and Dry Blood Sampling During Artemis Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), Stephanie Krueger is no longer with the study as a Co-Investigator. CodyGutierrez and Sara Bustos Lopez, both from JES Tech, have been added to the list of Co-Investigators (Ed., 3/11/25).		
COI Name (Institution):	Nelman-Gonzalez, Mayra B.A. (KBR/NASA Johnson Space Center) Mehta, Satish Ph.D. (JES Tech) Diak, Douglas Ph.D. (Advanced Enterprise Global Information Technology Solutions/NASA Johnson Space Center) Colorado, Audrie Ph.D. (JES Tech) Chouker, Alexander Ph.D. (Ludwig Maximilian University of Munich, Germany) Osterman, Andreas Ph.D. (Ludwig Maximilian University of Munich, Germany) Simpson, Richard Ph.D. (University of Arizona) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (University of Texas Medical Branch) Young, Millennia Ph.D. (NASA Johnson Space Center) Gutierrez, Cody B.S. (JES Tech) Bustos Lopez, Sara B.S. (JES Tech)		
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Task Description:	Homeostasis of the immune system is essential for health and well-being, including protection and recovery from infectious disease, tumor surveillance, and immunological memory. The immune system is one of the body's most complicated systems, consisting of a myriad of distinct cellular populations, all with unique regulatory or effector functions, operating in disparate locations throughout the body. Immune dysregulation has been well characterized following spaceflight, thought to be associated with the acute stresses of landing and re-adaptation. Recently, certain specific aspects of adaptive immunity (primarily T cell function and viral reactivation), have been found persistently dysregulated during long-duration spaceflight. Further, there is a low but defined incidence of infectious disease and consistent reactivation of latent herpes viruses during flight, with incidence of allergy and hypersensitivity symptoms that are consistently elevated. [Antihistamines remain the second most prescribed medication aboard the International Space Station (ISS).] There is also evidence suggesting a persistent low-level inflammation during spaceflight. As these parameters remain dysregulated during 6 months of orbital flight, they suggest an increased crew health risk during prolonged exploration-class deep-space missions. Saliva, essentially a filtrate of blood plasma, is now understood to be a biosample that can be collected in a noninvasive fashion, while containing suitable relative information regarding physiological status. We propose to use dry saliva collection techniques, already in use and validated on the ISS for cortisol measurements. Dry saliva is collected on a tiny "book" containing filter paper pages, and air-dried for room temperature storage. We propose a longitudinal repeated measures study design to assess the above-described aspects of immunity in astronauts during deep space Artemis missions. The survey will be broad and include measures of stress, inflammation, and immune status. The reactivation of herpes viruses or activity of persistent virus is an excellent quantifiable biomarker for clinically relevant immune compromise. The first planned mission is Artemis II through Artemis V; however, we propose that if successful, this technology represents a capability to derive in-flight samples from Artemis missions of increasing duration, including lunar surface operations. Dry saliva samples and dry blood spots will be collected before, during, and after spaceflight. Blood samples for other relevant immune surveillance assays are proposed pre- and post-flight. During the definition year, novel protocols for collection of blood and saliva during microgravity conditions, as well as deployment of the biosamples onto the specific books/cards, will be developed and validated. NOTE: For all Aims, in addition to dry saliva samples collected during flight, pre/postflight liquid blood and saliva samples will also be collected. These liquid samples will afford a correlation between dry and liquid saliva (pre/post) for subjects and allow a comparison of pre/post findings against the existing database for ISS/ground/clinical subjects.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. It is imperative to develop an immune countermeasure strategy that will preserve proper immune function in space and ensure the health of the astronauts. The clinical risks for exploration spaceflight are unknown. Prolonged durations, increased radiation, stress, and further limits on living conditions may synergize to produce incidence of clinical disease. This study will help ascertain, based on existing spaceflight and to be determined (TBD) clinical correlative data (this study), precisely what risks may manifest for exploration crewmembers. This information will help guide future countermeasures work and treatment options. The available storage mass for deep space Artemis/Gateway missions will be incredibly constrained. Conditioned, meaning frozen or chilled, storage of biosamples will not be available on the Orion capsule, nor on the Gateway Moon station. Blood will not be collected. This information is necessary to prepare for eventual Mars missions. Dry saliva is proposed as a method to allow biosample collection and storage, in a fashion compatible with the constraints of these vehicles, to allow monitoring of crew physiology.
Research Impact/Earth Benefits:	This information is necessary to interpret clinical risk for future prolonged deep space exploration and will guide the design of countermeasures associated with such missions.
Task Progress:	Initial study involved both short (~2 weeks) and long-term (1 year) stability studies which were completed. Samples were delivered to the participating laboratories (NASA Johnson Space Center, Immunology and Virology Lab; as well as Nutrition Biochemistry Lab, University of Arizona). Following the data analysis completion, target filter papers were selected with the official assay complement (validated) for the flight. In parallel, a collaborating laboratory in Oregon is assessing the stability of dry blood analytes via a Roche cobas analyzer. There has been a secondary short (~2 weeks) and long-term (1 year) stability study initiated, targeting an expanded list of analytes. The short stability study has been completed and sample analysis underway to be completed in FY25. Results will provide an expanded official assay complement for flight, that will complete full validation following the analysis of the 1-year study. The longer 1-year stability study has completed time points out to 3 months, with batch analysis to occur following the extraction of the 1-year time point at the end of FY25. The Institutional Review Board (IRB) approval have been obtained for this study during this study. The necessity of an international agreement to enable European Space Agency (ESA) participation is being assessed by NASA. Feasibility Assessment and Select for Flight are being planned. Finally, hardware packaging and crew training are in progress by the NASA Johnson Space Center (JSC) Artemis science board. Parabolic flight tests have been completed for validation of all collection protocols.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Articles in Peer-reviewed Journals	Diak DM, Crucian BE, Nelman-Gonzalez M, Mehta SK. "Saliva diagnostics in spaceflight virology studies-A review." Viruses. 2024 Dec 12;16(12):1909. https://doi.org/10.3390/v16121909 ; PMID: 39772216 PMCID: PMC11680219 , Dec-2024